# Tharun V. Puthanveettil

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**Human-Inspired Robot Learning | Multimodal Learning | Human–Robot Interaction**

## Research Focus

I want to build robots that understand the people they work with and keep learning from them over time. I take inspiration from how people learn: we build on experience, recognize what a new situation requires, identify gaps in what we know, and acquire and retain useful knowledge. I want to investigate how robot-learning architectures can accumulate and use contextual knowledge across tasks, people, and environments, so that acquiring one skill makes the next easier to learn. My work spans multimodal human understanding, learning from intervention, policy explainability and predictive models, with physical-robot, simulation and offline evaluations.

## Research Threads

### Understanding People

PRIOR WORK · 2021–2024

*How can a robot recognize what people are doing and what they mean from incomplete, multimodal evidence?*

- Designed GCN–transformer models for action recognition and fall detection from multi-view human pose. Under occlusion, fusing views raised F1 from 0.76 to 0.82 (graduate course project). PoseFusion
- Co-developed Track Anything Rapter, in which a person picks a target by click, box, image, or text and a drone follows it using foundation-model perception and visual servoing.
- Implemented an exploratory policy that learns from demonstration, speech, and gesture on the NatSGD dataset (Snehesh Shrestha, UMD). Whether gesture helps is not yet validated.

### Learning from People

PHYSICAL WORKFLOW; SIMULATION STUDIES

*How can robots improve from people's demonstrations and corrections without losing what they already do well?*

- Built a workflow in which people demonstrate tasks, take over when the robot fails, and have their corrections reviewed and added back into training before a new policy is rolled out. Evaluation remains manual. Overview
- Built a step-level control interface through which people and AI models operate the arm with the same small, safety-checked actions. Every step is recorded, so human and model runs can be compared and reused for training.
- Developing a portable egocentric interface that records human demonstrations through the same data pipeline (interface in development). Overview
- Compared teacher–student training and learned auxiliary heads in simulation, testing whether policies retain useful guidance when privileged training inputs are withdrawn.

### Policy Explainability and Representation

EXPERIMENTS IN PROGRESS

*What has a policy learned, and what shapes its actions?*

- Studied what drives learned policies using attention maps, Grad-CAM, input ablations, prompt comparisons, and self-supervised versus pretrained representations. These are diagnostics, not causal explanations. Overview
- Conducted controlled simulation studies using activation patching and counterfactual scene edits to test how information in policy representations affects actions. Compared these interventions with attribution diagnostics and behavioral tests.

### Predictive Models, Inverse Dynamics, and Transfer

OFFLINE AND SIMULATION STUDIES

*How can predicted outcomes and inverse dynamics support action learning on physical robots?*

- Retrained a published video-action model's action head on teleoperation data (offline evaluation completed). In simulation, trained image-based inverse dynamics and compared inferred-action policies with recorded-label and executed-command controls. Overview
- Fine-tuned a world-action model in simulation, assessing future-feature prediction, instruction grounding and failure signals separately from task success. Physical validation remains pending.

## Education

**Master of Engineering in Robotics, GPA: 3.9/4.0** — **Aug. 2022 – May 2024**
*University of Maryland, College Park* — *Maryland, USA*

**Selected coursework:** Advanced Techniques in Visual Learning and Recognition; Perception for Autonomous Robots; Planning for Autonomous Robots; Control of Robotic Systems; Fundamentals for Artificial Intelligence and Deep Learning Framework.

**B.Tech. in Electronics and Communication Engineering, GPA: 8.59/10.0** — **Jul. 2014 – Apr. 2018**
*Vellore Institute of Technology* — *India*

## Research and Professional Experience

**Robotics Engineer – Robot Learning and Deformable Manipulation** **Jul. 2024 – Present**
*Optimal — assigned to General Motors R&D* *Michigan, USA*

- Architected and implemented a modular ROS 2 platform for bimanual robot learning, from teleoperated demonstration and synchronized data capture to training, simulation, staged testing, and physical deployment. Shared interfaces let devices, learning methods, and simulators change without rebuilding the workflow.
- Developed human-correction and step-level control workflows, enabling comparable human and policy rollouts and analysis of failures and interventions.
- Built foundation-model services for task-conditioned detection, segmentation, tracking, and pose servoing. Reusing embeddings and propagating masks over time cut perception latency by about 35% and sustained 30–40 FPS.
- Developed force-aware bimanual manipulation of deformable linear objects using impedance control and force-torque feedback, incorporating elastic-tension regulation to support consistent manipulation.
- Built a calibrated digital twin and a behavior-tree-driven synthetic-data workflow to test how policies degrade under changes in sensing, timing, contact, and appearance.
- Designed the Cartesian motion-control and safety layer for learned policies, and supervised its extension to compliant (admittance-controlled) dual-arm data collection and rollout.
- Compared action-chunking and open-source vision-language-action policies while developing evaluation measures beyond success rate (partial progress, failure type, recovery, repeatability).

**Researcher – AI and Robotics** **Feb. 2023 – May 2024**
*CATT Laboratory, University of Maryland* *Maryland, USA*

- Architected the fusion pipeline for multimodal anomaly detection, with separate LiDAR, odometry, and network encoders and reconstruction-based scoring. The published system achieved up to 98% accuracy with vector-based reconstruction loss versus 72% with scalar-based loss, and 97% with half the training data.
- Co-developed Track Anything Rapter, a ROS 2 aerial system for multimodal target specification (see Research Threads).

**Volunteer Researcher – Multimodal Human–Robot Interaction** **Oct. 2022 – Jun. 2023**
*Perception and Robotics Group (PRG), University of Maryland* *Maryland, USA*

- Implemented an exploratory demonstration, speech, and gesture policy (see Research Threads).
- Implemented a 3D human body mesh reconstruction pipeline using the SMPL model.

**Researcher – Collaborative Robotics and Computer Vision** **May 2021 – Dec. 2021**
*Indian Institute of Science* *Bangalore, India*

- Developed a vision-guided precision-weeding robot that estimates plant roots and spread, then plans with hybrid inverse kinematics to reach weeds while reducing contact with crops. First-author work, ICSR + AI 2025.
- Built synthetic-data and augmentation pipelines for weed localization, using hard-negative sampling and structured distractors to improve robustness under challenging backgrounds.
- Co-designed a dual-purpose end effector (plucking plus spraying or watering) and adaptive position/torque control for delicate plants.
- Developed and evaluated a U-Net monocular depth model in an IROS 2021 challenge setting.

**Project Engineer – AI Innovation** **Jul. 2018 – Apr. 2021**
*CTO Office, Wipro Digital* *Bangalore, India*

- Led applied-research prototypes in computer vision, HCI, recommendation, generative modeling, and mobile inference, and coordinated quarterly prototype selection and delivery.
- Built tool tracking, mobile vehicle recognition, photogrammetry, human-pose estimation, virtual try-on, and recommendation components for industrial and retail workflows.

### Selected Earlier Technical Projects

**Implicit Neural Representation and SLIC Segmentation** | *PyTorch, TensorFlow, OpenCV* 2023

- Implemented coordinate-conditioned image outpainting and deep-feature superpixel segmentation, including controlled comparisons with simpler baselines.

**Maze Runner and Autonomous Retrieval Robot** | *ROS 2, Nav2, visual servoing* 2023

- Built visual-cue-based maze navigation in Gazebo and an onboard-perception retrieval robot using sensor fusion and visual servoing.

**Reinforcement-Learning Cricket Simulation** | *adaptive agents and simulation* 2020

- Built a simulated bowler agent that adapts to batting styles and field configurations. code

## Publications

*(Also published as V. P. Tharun.)*

1. A. Zoulkarni, M. Noorani, J. S. Baras, J. Mirenzi, **T. V. Puthanveettil**, and C. D. Grody. "Efficient Integration of Domain Knowledge and Data-Driven Methods for Securing Autonomous Cyber-Physical Systems." IEEE ROSE 2026. DOI: 10.1109/ROSE69181.2026.11570974.
2. **V. P. Tharun** and A. R. Chowdhury. "Vision Based Hybrid IK Task Planning with Feedforward Neural Network for Collaborative Plant-Robot Interaction in Precision Farming." *International Conference on Social Robotics + AI (ICSR + AI 2025)*, LNAI 16132, pp. 253–267. Springer, 2026. DOI: 10.1007/978-981-95-2382-5_18.
3. M. Noorani, **T. V. Puthanveettil**, A. Zoulkarni, J. Mirenzi, C. D. Grody, and J. S. Baras. "Multimodal Anomaly Detection for Autonomous Cyber-Physical Systems Empowering Real-World Evaluation." *GameSec 2024*, LNCS 14908, pp. 306–325. DOI: 10.1007/978-3-031-74835-6_15.
4. L. P. Pratap, P. M. Shailendrasingh, A. Anand, and **V. P. Tharun**. "Wall-Climbing Robot Using Soft Robotics." *ICPCSI*, 2017.

Earlier work in data mining and forecasting (2018–2020): five papers and reports on rainfall forecasting and water-quality assessment. Google Scholar

## Preprints and Research Outputs

1. **T. V. Puthanveettil**, A. Singh, Y. Jain, V. Bukka, and S. Arjun S. "Auto-Platoon: Freight by Example." *arXiv:2405.11659*, 2024. code
2. **T. V. Puthanveettil** and F. Obaid ur Rahman. "Track Anything Rapter (TAR)." *arXiv:2405.11655*, 2024. code + demos
3. **T. V. Puthanveettil**, A. Singh, and A. Balram. "Pose Fusion - Multi-View Pose Integration for Comprehensive Action Recognition." UMD graduate course project. Project poster and preprint. preprint
4. **V. P. Tharun** et al. "Application of Mobile Collaborative Robot using Deep Learning in Precision Weed Control." Manuscript, 2021.
5. **V. P. Tharun** et al. "Event-Based Dynamic Obstacle Avoidance in Outdoor Environments." IROS 2021 workshop poster.

## Technical Skills

**Robot learning:** PyTorch, TensorFlow, LeRobot, action-chunking policies, vision-language-action models, diffusion/flow policies, imitation learning, self-supervised learning

**Robotics and control:** ROS 2, ROS, MoveIt 2, impedance/compliance control, visual servoing, behavior trees, bimanual manipulation, force-torque feedback

**Simulation and perception:** Isaac Sim/Lab, MuJoCo, Gazebo, OpenCV, Open3D, RGB-D cameras, detection, segmentation, tracking, pose estimation

**Programming and systems:** Python, C++, SQL, Linux, Docker, Git, REST APIs

## Intellectual Property, Mentoring, and Communication

**Patents:** Inventor on 11 invention disclosures at General Motors, two on the patent route and three protected as trade secrets; details confidential.

**Research supervision:** Supervised an engineering intern extending the robot-learning platform for compliant bimanual data collection, learning, analysis, and rollout.

**FIRST Robotics:** Electrical Programming Head Coach, The Irrational Engineers (Jun. 2024–Present); mentor high-school students in controls, computer vision, and software.

**Writing:** *Human in the Loom*, essays on robot learning, embodied intelligence, and credible research practice. Essay archive